Experiment – 7:

Continuous-Time Fourier Transform and Spectrum

OBJECTIVES

(i) To analyze the continuous-time Fourier transform of different signals and interpret their

corresponding frequency-domain characteristics.

(ii) To determine the magnitude and phase spectra of continuous-time signals and analyze their

significant spectral components.

(iii) To verify important Fourier transform properties and examine their effects on the

corresponding time-domain and frequency-domain representations.

(iv) To analyze signal bandwidth and spectral components and relate the frequency-domain characteristics to communication system applications.

(2) To determine the magnitude and phase spectra of continuous-time signals and

analyze their significant spectral components.

Experimental Procedure

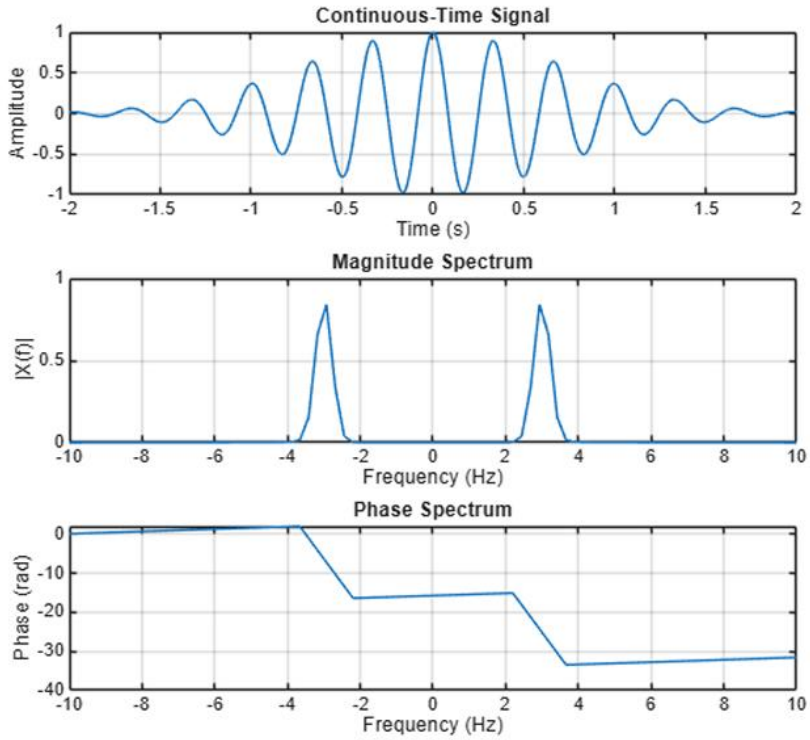
- Define t and the sampling interval.
- Generate x(t)=e^(−t2)cos(2π3t).
- Compute the FFT-based transform with N = 4096.
- Plot the signal, magnitude and unwrapped phase.
- Identify the principal spectral peaks.

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```
clc; clear; close all;
%% 7(ii) Determination of Magnitude and Phase Spectra of Continuous-Time Signals
figure;
% Magnitude and Phase Spectrum of a CT Signal
t = -2:0.001:2;
dt = t(2)-t(1);
x = exp(-t.^2).*cos(2*pi*3*t);
N = 4096;
X = fftshift(fft(x,N))*dt;
f = (-N/2:N/2-1)/(N*dt);
magX = abs(X);
phaseX = unwrap(angle(X));
subplot(3,1,1);
plot(t,x); grid on;
xlabel('Time (s)'); ylabel('Amplitude');
title('Continuous-Time Signal');
subplot(3,1,2);
plot(f,magX); xlim([-10 10]); grid on;
xlabel('Frequency (Hz)'); ylabel('|X(f)|');
title('Magnitude Spectrum');
subplot(3,1,3);
plot(f,phaseX); xlim([-10 10]); grid on;
xlabel('Frequency (Hz)'); ylabel('Phase (rad)');
title('Phase Spectrum');
```



(iii) Verification of Fourier Transform Properties and Frequency-Domain Characteristics

Experimental Procedure

• Generate x(t)=e^(−t2).

• Create x(t−0.4) using interpolation.

• Compute the FFT of the shifted signal.

• Form the theoretical transform using the time-shift property.

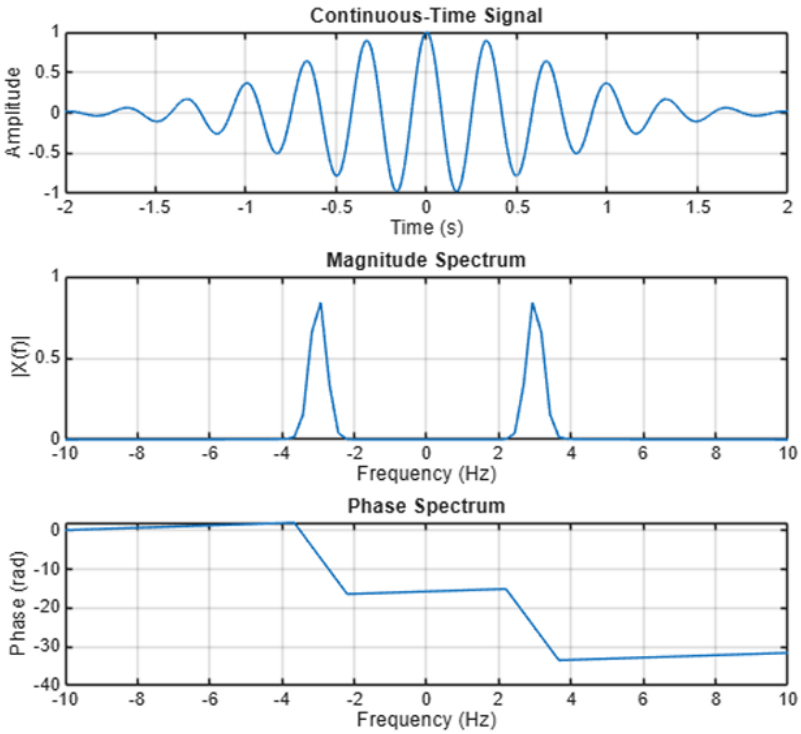
• Compare computed and theoretical magnitude and phase.

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```
1   clc; clear; close all;
2   %% 7(ii) Determination of Magnitude and Phase Spectra of Continuous-Time Signals
3   figure;
4   % Magnitude and Phase Spectrum of a CT Signal
5   t = -2:0.001:2;
6   dt = t(2)-t(1);
7   x = exp(-t.^2).*cos(2*pi*3*t);
8   N = 4096;
9   X = fftshift(fft(x,N))*dt;
10  f = (-N/2:N/2-1)/(N*dt);
11  magX = abs(X);
12  phaseX = unwrap(angle(X));
13  subplot(3,1,1);
14  plot(t,x); grid on;
15  xlabel('Time (s)'); ylabel('Amplitude');
16  title('Continuous-Time Signal');
17  subplot(3,1,2);
18  plot(f,magX); xlim([-10 10]); grid on;
19  xlabel('Frequency (Hz)'); ylabel('|X(f)|');
20  title('Magnitude Spectrum');
21  subplot(3,1,3);
22  plot(f,phaseX); xlim([-10 10]); grid on;
23  xlabel('Frequency (Hz)'); ylabel('Phase (rad)');
24  title('Phase Spectrum');
```


Continuous-Time Signal
Amplitude
Time (s)
Magnitude Spectrum
|X(f)|
Frequency (Hz)
Phase Spectrum
Phase (rad)
Frequency (Hz)